



The University of Manchester

THE DEPARTMENT OF PHYSICS AND ASTRONOMY

FIRST SEMESTER REPORT

Developing discrimination techniques for $HH \rightarrow bb\tau_h\tau_l$ events at ATLAS

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December 2019

MPhys Project

Abstract

A method of discriminating $HH \rightarrow bb\tau_h\tau_l$ signal events from background at the ATLAS experiment is reviewed. It uses a discriminator ψ that combines the techniques of b -jet re-scaling, angular separation selection of b -jets and taus, and di-leptonic $t\bar{t} \rightarrow bbl$ event reconstruction. The penalty discriminator ψ_T used for event reconstruction of dileptonic $t\bar{t}$ decays was then investigated in an attempt to further reduce the $t\bar{t}$ background contribution to di-Higgs signals. Two new discriminators $\psi_T^{(2)}$ and $\psi_T^{(3)}$ were produced and used in conjunction with ψ_T to select di-leptonic $t\bar{t}$ events.

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1 Introduction

The Standard Model (SM) of particle physics provides a remarkably accurate description of experimental results from particle physics experiments [1]. It is the most well tested theory of particle physics that describes the interactions between elementary particles[2].

One crucial aspect of the Standard Model is the Higgs mechanism. This mechanism describes the generation of masses for known elementary particles via electroweak symmetry breaking [3]. It predicts a scalar (spin 0) Higgs field which manifests itself as a physical Higgs particle H when excited [4].

In 2012, ATLAS and CMS collaborations confirmed the existence of the neutral Higgs particle and measured its mass to be around 125 GeV [5-6]. Since then, experiments had studied the various properties of the Higgs boson, making rates and differential measurements of its production and decay modes [7-10]. So far, all results agrees with the SM predictions within experimental and theoretical uncertainties [11].

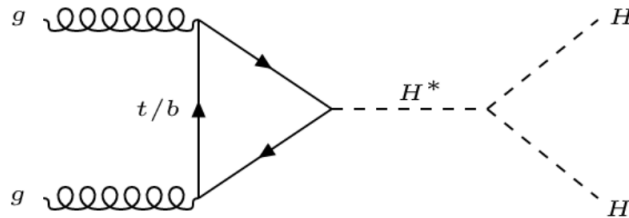


Figure 1: Feynman diagram for trilinear Higgs self-coupling

One of the future experiments planned at the Large Hadron Collider (LHC) involves the measurement of the trilinear Higgs boson self coupling constant λ_3 [12]. The Higgs mechanism predicts that Higgs field can self-interact, so a Higgs boson can further decay two Higgs bosons, as shown in Figure 1. It is thus important to study di-Higgs decays as it gives access to the trilinear self coupling constant. Efforts to measure the various decay channels of di-Higgs production is an active field. Examples include the channels of $HH \rightarrow b\bar{b}b\bar{b}$, $HH \rightarrow b\bar{b}b\bar{b}$, $HH \rightarrow b\bar{b}q\bar{q}l\nu$, $HH \rightarrow b\bar{b}\tau\tau$ and $HH \rightarrow b\bar{b}\gamma\gamma$ [13-14].

In this report, the di-Higgs decay channel $HH \rightarrow b\bar{b}\tau_h\tau_l$ was chosen as the subject of study. The Feynman diagram describing this decay channel is shown in Figure 2. One of the Higgs particle decays into $b\bar{b}$ quarks which form two hadronic b-jets that are detected by the ATLAS detector. The other Higgs particle first decays into two τ leptons, with one of them further decaying hadronically (τ_h) and the other leptonically (τ_l). The τ_h lepton produces two jets that can be detected and the τ_l produces a light lepton (e or μ) that is also detectable. The decay of the two $\tau_h\tau_l$ leptons also produce neutrinos whose presence is not directly detected but inferred from the missing transverse energy E_T^{miss} of the final products. This decay channel is preferable to study because of the relatively large branching ratios of $H \rightarrow b\bar{b}$ (58%) and $H \rightarrow \tau\tau$ (6.3%) compared to other decay channels.

One of the main background event for $HH \rightarrow b\bar{b}\tau_h\tau_l$ decays is dileptonic $t\bar{t}$ decay into two $b\bar{b}$ quarks, a hadronic τ_h and leptonic τ_l ($t\bar{t} \rightarrow b\bar{b}\tau_h\tau_l$) [15]. This dileptonic $t\bar{t}$ decay mode produces similar final state products as the di-Higgs decay mode we wish to study and thus deserve

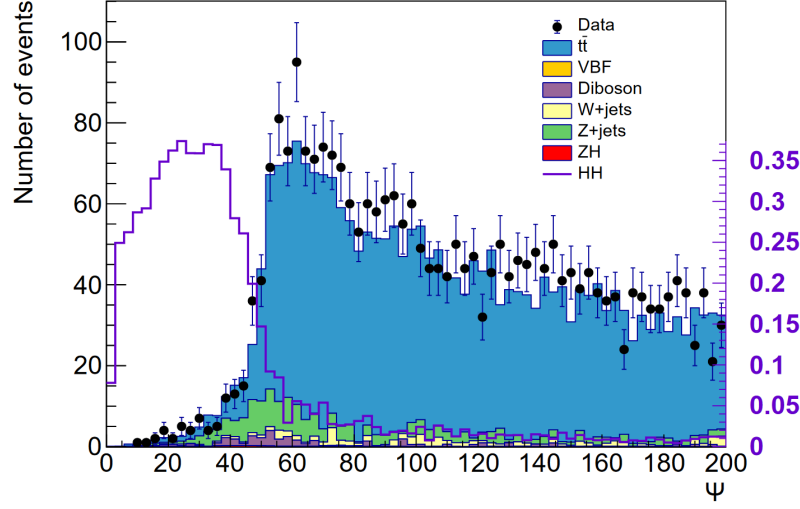


Figure 3: Histogram of events analysed using the ψ discriminator. The right axis correspond to the di-Higgs signal events. Note the rarity of signal events compared to background, especially the $t\bar{t}$ background.

neutrinos sum up to E_T^{miss} recorded by the detector. ψ_ϵ measures how much the b -jets momenta need to be re-scaled to fit the kinematic criteria. The re-scaling of b -jets momenta is necessary since energy reconstruction of b -jets can be poor within the ATLAS detector. The idea is that smaller re-scaling constants corresponds to a greater likelihood that the observed event is a di-Higgs decay. $\psi_{\Delta R}$ measures how well the angular separations between the two b -jets and between the two $\tau_h\tau_l$ leptons agree with predictions. In theory, the pairs of b -jets and $\tau_h\tau_l$ leptons have a large mass difference from the Higgs particle which they decayed from and so should be strongly boosted, resulting in small angular separations for each pair. ψ_{T} measures how likely the observed event is a dileptonic $t\bar{t}$ decay event. If the event kinematically reconstructs well to a $t\bar{t}$ event, it is less likely to be a di-Higgs signal and more likely to be a $t\bar{t}$ background.

The six discriminants ψ_{bb} , $\psi_{\tau\tau}$, $\psi_{E_T^{miss}}$, ψ_ϵ , $\psi_{\Delta R}$ and ψ_T thus measures different aspects of an observed event and matches them to the theory of di-Higgs decay. Combining them in a quadrature manner gives us the ψ discriminator which describes overall how likely an observed event is di-Higgs signal. Detailed formulation for each of the six discriminant can be found in [17,18].

The results achieved by the previous pair of students using the ψ discriminant is shown in Figure 3. The shapes of the signal and combined background showed that the ψ discriminator was successful in separating signal and background events. Signal events mostly have small ψ values while background events tend to have higher ψ values. It can also be seen that $t\bar{t}$ events were the most dominant background.

2.1 A detailed look at the ψ_T discriminant

While Cookman & Harris were successful in using the ψ discriminator to select di-Higgs signal events, they stated in their report's conclusion that their formulation of ψ_T had been relatively simplistic and that a more sophisticated formulation of ψ_T will be of benefit to the analysis of di-Higgs decay. In particular, some signal events were shown to have large absolute values of ψ_T , which was undesirable. In this subsection we discuss how the previous students formulated the ψ_T discriminant and its shortcomings.

The dileptonic $t\bar{t}$ background Cookman & Harris considered was $t\bar{t} \rightarrow b\bar{b}\tau_h l$. One t quark decays into a hadronically decaying τ_h and the other t quark decays into a light lepton l . This background event produces similar final state products as that of the di-Higgs signal. This is the most dominant background to the di-Higgs signal. The previous pair of students hence formulated ψ_T to select this $t\bar{t}$ decay mode. Note that a less dominant $t\bar{t}$ background event is $t\bar{t} \rightarrow b\bar{b}\tau_h \tau_l$, shown in Figure 4. ψ_T does not describe this process.

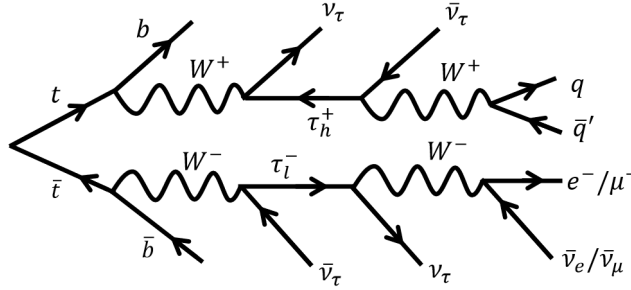


Figure 4: Feynman diagram for $t\bar{t} \rightarrow b\bar{b}\tau_h \tau_l$

The $t\bar{t} \rightarrow b\bar{b}\tau_h l$ event produces three neutrinos: ν_l produced with the light lepton, ν_τ^1 produced with the τ_h lepton and ν_τ^2 produced from decay of the τ_h . The presence of three neutrinos make event reconstruction difficult. To simplify the problem, Cookman & Harris 'combined' the ν_τ^1 and ν_τ^2 into a single composite neutrino ν_τ^* for analysis. This method is justifiable since in high-energy collisions particles are strongly boosted so that their motions are very close to being collinear. Event reconstruction of $t\bar{t} \rightarrow b\bar{b}\tau_h l$ decays then only involve two neutrinos which is easy to handle.

The event reconstruction for $t\bar{t} \rightarrow b\bar{b}\tau_h l$ decays is then achieved by varying the momenta of ν_τ^* and ν_l to search for the global minimum of a penalty function Δ_T defined as:

$$\Delta_T^2 = (m_t - M_t)^2 + (m_{\bar{t}} - M_{\bar{t}})^2 + (m_{W_l} - M_W)^2 + (m_{W_{\tau_h}} - M_W)^2 + (p_{\nu_l}^x + p_{\nu_\tau^*}^x - E_T^{miss^x})^2 + (p_{\nu_l}^y + p_{\nu_\tau^*}^y - E_T^{miss^y})^2, \quad (2)$$

where m_i denotes the reconstructed invariant mass of particle i and M_i denotes the known invariant mass of particle i . While p_i^x and p_i^y is the x and y components of the i particle's momentum measured in the detector's transverse plane. E_T^{miss} denotes the event's measured missing transverse energy as usual.

The penalty function Δ_T is essentially a function of the two neutrino's three-momenta $\vec{p}_{\nu_\tau^*}$ and $\vec{p}_{\nu_l}$. It measures how kinematically consistent an observed event with appropriate final state products is with the $t\bar{t} \rightarrow b\bar{b}\tau_h l$ decay mode.

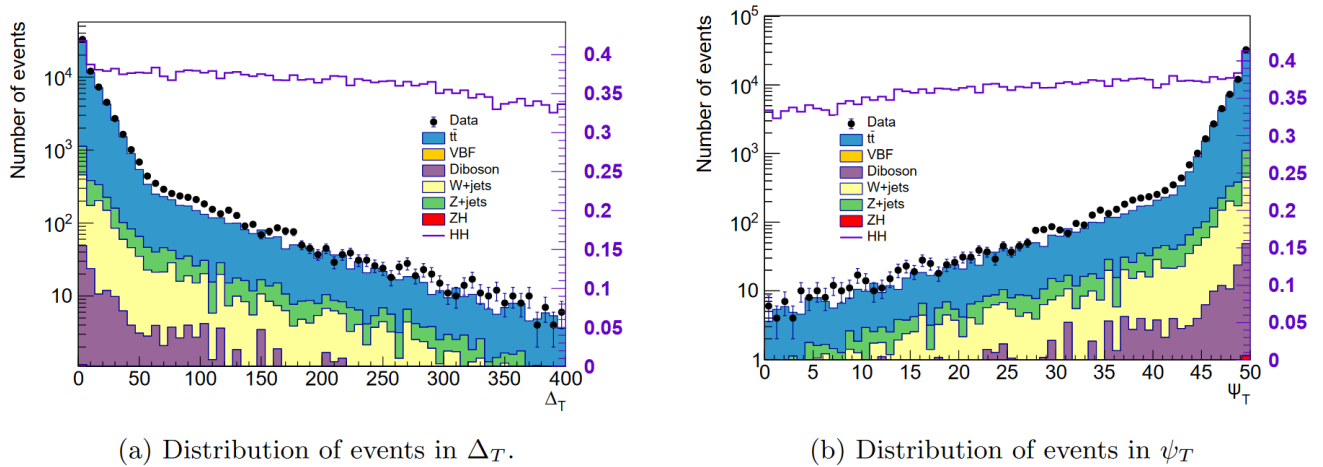


Figure 5: (a) Histogram of events analysed using the Δ_T penalty function. Note that there exists $t\bar{t}$ events with a large value of Δ_T . (b) Histogram of events analysed using the ψ_T discriminator. We can see that for small values of ψ_T , number the background events is at its lowest.

ψ_T is then defined in terms of the penalty function Δ_T :

$$\psi_T = \frac{50}{400}(400 - \Delta_T).$$

The numerical coefficients are introduced for scaling purposes so that ψ_T fits well with the other five ψ_i discriminants. This form of ψ_T ensures that events with a low value of Δ_T , signalling that they are likely $t\bar{t}$ events, produce a high value of ψ_T . Conversely, events with a high value of Δ_T are assigned a low value of ψ_T . This means that events that are not likely to be $t\bar{t}$ background are treated as more likely to be di-Higgs signal.

The results obtained by Cookman & Harris using the Δ_T function and the ψ_T discriminator on dileptonic $t\bar{t}$ events is shown in Figure 5. It can be seen background events diminish for large values of Δ_T and small values of ψ_T as expected. The relatively constant number of di-Higgs signal events across the range of Δ_T and ψ_T showed that these two discriminants did not have a bias when selecting di-Higgs signals. However, the fact that some $t\bar{t}$ events had abnormally large values of Δ_T and some di-Higgs signals had low values of Δ_T showed that this method was not perfect. The discriminatory power of Δ_T and hence ψ_T can still be improved.

Cookman & Harris' formulation of ψ_T is after all just a simplistic version that only considered dileptonic $t\bar{t} \rightarrow b\bar{b}\tau_h l$ background. The assumption of the composite neutrino was also a crude approximation. The $t\bar{t}$ background event is $t\bar{t} \rightarrow b\bar{b}\tau_h \tau_l$ was also not considered due to the increased complexity brought forward by the presence of additional neutrinos from decay of τ_l . The general idea is then continue the work of the previous students and to improve the discriminatory power of ψ_T by investigating the dileptonic $t\psi t$ decay process in more detail and sophistication. This can include investigating how ψ_T can be modified to include other dileptonic $t\bar{t}$ events on top of $t\bar{t} \rightarrow b\bar{b}\tau_h l$. We also aim to improve on the event reconstruction method so that it can handle the presence of more neutrinos. Our attempts to do so is described in the next section.

3 Improving the ψ_T discriminant

To improve the ψ_T discriminant so that it can better select dileptonic $t\bar{t}$ events, we investigated the three following $t\bar{t}$ channels :

- $t\bar{t} \rightarrow b\bar{b}WW \rightarrow b\bar{b}ll\nu_l\nu_l$
- $t\bar{t} \rightarrow b\bar{b}WW \rightarrow b\bar{b}l\tau\nu_l\nu_\tau \rightarrow b\bar{b}ll_2\nu_l\nu_\tau\nu_{\tau_2}\nu_{l_2}$
- $t\bar{t} \rightarrow b\bar{b}WW \rightarrow b\bar{b}\tau\tau\nu_\tau^{(1)}\nu_\tau^{(2)} \rightarrow b\bar{b}ll_2\nu_l\nu_{l_2}\nu_\tau^{(1)}\nu_\tau^{(2)}\nu_{\tau_2}^{(1)}\nu_{\tau_2}^{(2)}$

The first decay channel ($t\bar{t}1$) is the double prompt W decays to light leptons . Each W boson from the decay of $t\bar{t}$ decays into a light lepton l and a neutrino ν_l . For the second decay channel ($t\bar{t}2$), one W boson decays promptly to a light lepton l and the other to a τ lepton. The τ lepton then decays to light lepton l_2 and produces neutrinos $\nu_{\tau_2}\nu_{l_2}$. For the third decay channel ($t\bar{t}3$), both W bosons decay to τ leptons which then decay to two light leptons (l and l_2) plus neutrinos. The number of neutrinos present in the final state products for each decay channel is 2, 4 and 6 respectively. Kinematically reconstructing the momenta for these neutrinos hence became the main focus of our work in the first semester.

For the $t\bar{t}1$ decay channel with only 2 neutrinos in the final state products , the form of Δ_T as shown in Equation 2 can be readily used, since it is just a standard dilepton $t\bar{t}$ decay. For the $t\bar{t}2$ and $t\bar{t}3$ decay channels, two new penalty functions Δ_{T2} and Δ_{T3} were created to account for the extra neutrinos. That means that Δ_T will be used to select $t\bar{t}1$ events, Δ_{T2} will be used to select $t\bar{t}2$ events and Δ_{T3} will be used to select $t\bar{t}3$ events. The three penalty functions can then be used in conjunction to select dileptonic $t\bar{t}$ events that produce two light leptons as their final decay products.

3.1 The Δ_{T2} penalty function

The Δ_{T2} penalty function is used to select $t\bar{t}2$ events. The decay process of $t\bar{t}2$ occurs in three stages. In the first stage, the $t\bar{t}$ pair decays into a $b\bar{b}$ pair and a pair of oppositely charged W bosons. In the second stage, the oppositely W bosons further decay leptonically to produce the particle $\tau\nu_\tau l\nu_l$. In the third stage, the unstable τ particle decays into two neutrinos $\nu_{\tau_2}\nu_{l_2}$ and a light lepton l_2 . The final state products are then $b\bar{b}ll_2\nu_l\nu_\tau\nu_{\tau_2}\nu_{l_2}$.

Since the momenta of the particles $b\bar{b}ll_2$ is detectable, our task is to 'guess' the unknown momenta of the four neutrinos $\nu_l\nu_\tau\nu_{\tau_2}\nu_{l_2}$ to reconstruct $t\bar{t}2$ events. To facilitate this 'guessing', we use an approximation that the three-momenta of neutrinos $\nu_{\tau_2}\nu_{l_2}$ are in the same direction as that of light lepton l_2 . This follows a similar argument presented in Section 2.1. l_2 , ν_{τ_2} and ν_{l_2} are strongly boosted such that they are very close to being collinear. We can then treat the two neutrinos ν_{τ_2} and ν_{l_2} together in our analysis.

The combined three-momenta of ν_{τ_2} and ν_{l_2} can thus be defined in terms of the measured momentum of l_2 :

$$\begin{aligned}\vec{p}_{\nu_{\tau_2}\nu_{l_2}} &= \beta\vec{p}_{l_2}, \\ \vec{p}_{\nu_\tau\nu_e} &= \beta\vec{p}_e\end{aligned}\tag{4}$$

where β is a scaling constant.

The exact form of Δ_{T2} can now be written as:

$$\Delta_{T2}^2 = (m_t - M_t)^2 + (m_{\bar{t}} - M_{\bar{t}})^2 + (m_{W_l} - M_W)^2 + (m_{W_\tau} - M_W)^2 + (\beta p_{l_2}^x + p_{\nu_l}^x + p_{\nu_\tau}^x - E_T^{miss^x})^2 + (\beta p_{l_2}^y + p_{\nu_l}^y + p_{\nu_\tau}^y - E_T^{miss^y})^2, \quad (1)$$

The Δ_{T2} penalty function was tested for its efficiency by using it to analyse a Monte-Carlo (MC) data set of dileptonic $t\bar{t}$ events. The method was to first preselect events with at least 2 high p_T (> 20 GeV) b -jets, 2 high p_T (> 20 GeV) leptons and then use Minuit2 numerical minimization to find a global minimum of Δ_{T2} for events that passed the selection cuts. For each event, the minimization was performed 100 times. At the start of each minimization, the three momenta $p_{\nu_\tau}^i$ and $p_{\nu_l}^i$ of neutrinos ν_τ and ν_l was randomly initialised to a starting value between $[-2p_l^i, 2p_l^i]$, where p_l^i is the momentum of the lepton l from prompt W decay. While β was initialised to a starting value between $[0, 10]$.

3.2 The Δ_{T3} penalty function

The Δ_{T3} penalty function is used to select $t\bar{t}3$ events. The decay process of $t\bar{t}$ involves two τ leptons that further decay leptonically to form two light leptons. The final state products are $b\bar{b}l l_2 \nu_l \nu_{l_2} \nu_\tau^{(1)} \nu_\tau^{(2)} \nu_{\tau_2}^{(1)} \nu_{\tau_2}^{(2)}$.

As before, the three-momenta of neutrino pairs $\nu_l \nu_{\tau_2}^{(1)}$ and $\nu_{l_2} \nu_{\tau_2}^{(2)}$ are approximated to be in the same direction as that of light leptons l and l_2 respectively. The three-momenta of each neutrino pair is thus combined and defined as:

$$\vec{p}_{\nu_l \nu_{\tau_2}^{(1)}} = \beta_1 \vec{p}_l, \quad (5)$$

$$\vec{p}_{\nu_{l_2} \nu_{\tau_2}^{(2)}} = \beta_2 \vec{p}_{l_2}, \quad (6)$$

where β_1 & β_2 are scaling constants.

The exact form of Δ_{T3} is then defined as:

$$\Delta_{T3}^2 = (m_t - M_t)^2 + (m_{\bar{t}} - M_{\bar{t}})^2 + (m_{W_\tau} - M_W)^2 + (m_{W_\tau} - M_W)^2 + (\beta_1 p_l^x + \beta_2 p_{l_2}^x + p_{\nu_\tau^{(1)}}^x + p_{\nu_\tau^{(2)}}^x - E_T^{miss^x})^2 + (\beta_1 p_l^y + \beta_2 p_{l_2}^y + p_{\nu_\tau^{(1)}}^y + p_{\nu_\tau^{(2)}}^y - E_T^{miss^y})^2. \quad (2)$$

The Δ_{T3} function is then used to analyse the same MC set of dileptonic $t\bar{t}$ events. The pre-selection criteria is the same as described for $t\bar{t}2$ events. The minimization was also performed 100 times for each event that passed the selection cuts.

3.3 Results and Discussion

The results of using the Δ_T , Δ_{T2} and Δ_{T3} penalty functions on the MC dataset of dileptonic $t\bar{t}$ decays is summarised in Table 1. Table 1 shows that the 71% of $t\bar{t}$ MC events that passed selection cuts have a low Δ_T value of less than 400. When the Δ_{T2} discriminant was used, 92% of preselected events had $\Delta_{T2} < 400$. For Δ_{T3} , 94% of preselected events had $\Delta_{T3} < 400$. The high amount of events passing the $\Delta_{T2} < 400$ (92%) and $\Delta_{T3} < 400$ (94%) selection was found to be due to the β scaling constants converging to close to 0 in most minimizations. This means that $\Delta_{T2} < 400$ is able to select both $t\bar{t}1$ and $t\bar{t}2$ events, since a zero β factor would

Table 1: Experimental Results

	No. of events with $\Delta_{T_i} < 400$
Δ_T	71%
Δ_{T2}	92%
Δ_{T3}	94%

mean that the neutrino pair $\nu_{\tau_2}\nu_{l_2}$ has zero momentum, reducing the problem to a simple $t\bar{t}1$ decay. Conversely, the Δ_{T3} penalty function is able to select $t\bar{t}1$ and $t\bar{t}2$ events since either of the constants β_1 and β_2 are allowed to be zero.

To assess whether these obtained values are consistent with the events present within the MC data set, we first used branching ratios to calculate the decay probability for each of the three channels. The $t\bar{t}1$ channel was found to have a branching ratio of 73.5%. While the $t\bar{t}2$ and $t\bar{t}3$ channels had branching ratios of 24% and 2.5% respectively. The experimental values obtained in Table 1 thus seem to agree well theory. For example, since Δ_{T2} selects both $t\bar{t}1$ and $t\bar{t}2$ events, it was expected to select $73.5\% + 24\% = 97.5\%$ of the dileptonic $t\bar{t}$ events. In actual fact, it had an efficiency of 92%, which is close.

However, although the results agree with branching ratio calculations, it is more important that they agree with the actual event distribution generated by the MC dataset, since the MC event distribution does not necessarily follow the branching ratios. Further analysis of the MC dataset used in this report showed that the MC event distribution was actually 86% for $t\bar{t}1$ events, 14% for $t\bar{t}2$ events and 1% for $t\bar{t}3$ events. This was done using the 'leptruthMatched' value assigned to each MC event. This value that ranges between 0 and 2 (for this MC dataset) tells us whether a MC event is $t\bar{t}1$, $t\bar{t}2$ or $t\bar{t}3$. Additional analysis using this value also showed that the 71% of events selected by Δ_T contained a significant amount of false positives of $t\bar{t}2$ and $t\bar{t}3$ events. The same is true for Δ_{T2} and Δ_{T3} . The next immediate step for us to do thus is to eliminate these false positives.

4 Future work plan

It can be seen from the previous section that the Δ_{T2} and Δ_{T3} can still be improved to more accurately select dileptonic $t\bar{t}$ events. It will be of immediate benefit to continue improving Δ_{T2} and Δ_{T3} for greater discriminatory power.

The workplan for the next semester is to finally look at dileptonic $t\bar{t}$ events where a τ_h lepton decays hadronically and a τ_l lepton decays leptonically. Up till now, we had only been analysing dileptonic $t\bar{t}$ events that do not consist of a hadronically decaying τ_h lepton. We also need to test the use of Δ_{T2} and Δ_{T3} on other datasets, which are the di-Higgs data and other di-Higgs background MC, to see if our penalty functions are reliable in selecting only $t\bar{t}$ events. We would also have to work on formulating a new ψ_T discriminator using the new penalty functions Δ_{T2} and Δ_{T3} created in the first semester. Thereafter we would be able to combine the ψ_T discriminator or with the five other discriminators to see how well the new combined ψ discriminator perform in discriminating di-Higgs signal-background.

5 Conclusion

This report concludes with a summary of the work we performed in this first semester.

In this semester, we reviewed the method used by Cookman & Harris in discriminating $HH \rightarrow b\bar{b}\tau_h\tau_l$ signal-background, which was the usage of the ψ discriminator. We then proceeded to improve on the ψ_T discriminator since the formulation by the previous pair of students was simplistic and only considered a single $t\bar{t} \rightarrow b\bar{b}\tau_h l$ decay channel. We first chose to study dileptonic $t\bar{t}$ decays with two light leptons as decay products and came up with the two new penalty functions Δ_{T2} and Δ_{T3} to select these events. Our experimental results show that our method is efficient to a certain extent but should still be improved. Nevertheless, we made satisfactory progress this semester in $t\bar{t}$ event reconstruction and look forward to proceeding to analyse di-Higgs signal in the next semester.

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